EXPERIMENT NO.: 1

DATE: 26-12-2025

EXPERIMENT TITLE:

SDLC-Based Requirement Analysis, Planning, Estimation, and Feasibility Study for a Travel Planning System

AIM:

To apply Software Development Life Cycle (SDLC) principles such as requirement analysis, planning, cost estimation, scheduling, and feasibility study to design a systematic and economical group travel plan**.**

PROBLEM DESCRIPTION::

The problem involves planning a group travel system for 10 members, including 7 non-vegetarians and 3 vegetarians, within a total duration of 7 days. The travel plan includes a stay in Darjeeling and must cover important tourist locations such as Tiger Hill, Batasia Loop, Ghoom Monastery, Peace Pagoda, Himalayan Mountaineering Institute, Padmaja Naidu Zoological Park, Tea Estates, and Mall Road. The plan must remain economical by using shared taxis, affordable accommodation, and cost-efficient food planning while satisfying all given constraints.

GIVEN CONDITIONS / CONSTRAINTS:

- Group size: 10 members (7 non-veg, 3 veg)
- Total duration: 7 days
- Stay in Darjeeling: 4 days and 3 nights
- Fixed tourist locations must be covered
- Local travel by shared taxis and walking only
- Affordable accommodation selection
- Separate food planning for vegetarian and non-vegetarian members

APPROACH / METHODOLOGY:

The travel plan was developed using a structured SDLC approach. Initially, all requirements and constraints were identified. The destination was analyzed for feasibility based on geography and accessibility. Travel, accommodation, food, and sightseeing plans were prepared systematically. Cost estimation was carried

out for each component, and feasibility analysis was performed to ensure that the plan is technically, economically, and operationally viable.

PLANNING AND ESTIMATION:

Transportation cost:

- Train travel (round trip): ₹20,000
- Local shared taxis: ₹5,000
- Total transportation cost: ₹25,000

Accommodation cost:

- ₹2,000 per room × 5 rooms × 3 nights = ₹30,000

Food expenses:

- ₹500 per person per day × 10 members × 4 days = ₹20,000

Entry fees and miscellaneous expenses: ₹5,000

Total Estimated Cost: ₹80,000

Per Person Cost: ₹8,000

SCHEDULING/ DESIGN CONSIDERATIONS:

Day 1: Travel from home city to New Jalpaiguri (NJP), proceed to Darjeeling, hotel check-in and rest

Day 2: Visit Tiger Hill, Batasia Loop, Ghoom Monastery, Tea Estates, and Mall Road

Day 3: Visit Himalayan Mountaineering Institute, Padmaja Naidu Zoological Park, Peace Pagoda, and local shopping

Day 4: Hotel check-out and return journey via NJP

Days 5–7: Reserved as buffer days for long-distance travel, rest, and contingencies

This schedule ensures efficient time utilization and complete coverage of required tourist locations.

FEASIBILITY ANALYSIS:

Technical Feasibility: The plan uses commonly available transportation and accommodation facilities.

Economic Feasibility: The total cost is economical and affordable for all members.

Operational Feasibility: Shared taxis and walking make the plan practical and easy to execute.

Schedule Feasibility: All activities fit within the 7-day duration without time overruns.

RESULT:

A complete SDLC-based travel planning system was successfully developed. The itinerary, cost estimation, and scheduling demonstrate that the proposed plan is systematic, economical, and feasible.

CONCLUSION:

This experiment proves that SDLC principles such as requirement analysis, planning, estimation, scheduling, and feasibility study can be effectively applied to real-world scenarios like travel planning. A structured SDLC approach results in better organization, optimal resource utilization, reduced risks, and a practical solution.